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### Impact tool

#### Abstract

An impact tool includes an output shaft for outputting torque, where the tightening torque of the output shaft to a workpiece is greater than or equal to 700 foot-pounds; and an impact assembly that provides an impact force to the output shaft. A main shaft ball groove includes a first ball groove that extends spirally about a main shaft axis and is concave on an outer surface and a second ball groove that extends spirally about the main shaft axis and is concave on the outer surface. An impact ball groove includes a third ball groove that mates with the first ball groove to accommodate the rolling ball and a fourth ball groove that mates with the second ball groove to accommodate the rolling ball. The included angle  $\alpha$  between the first ball groove and the second ball groove is less than the included angle  $\beta$  between the third ball groove and the fourth ball groove.

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## Background/Summary

### RELATED APPLICATION INFORMATION

(1) This application claims the benefit under 35 U.S.C. § 119 (a) of Chinese Patent Application No. 202310489252.4, filed on Apr. 28, 2023, Chinese Patent Application No. 202310493577.X, filed on Apr. 28, 2023, Chinese Patent Application No. 202321025625.4, filed on Apr. 28, 2023, Chinese Patent Application No. 202321021084.8, filed on Apr. 28, 2023, and Chinese Patent Application No. 202311019899.7, filed on Aug. 11, 2023, which applications are incorporated herein by reference in their entireties.

### TECHNICAL FIELD

(2) The present application relates to the technical field of power tools and, in particular, to an impact tool.

### BACKGROUND

(3) An impact tool can output rotary motion with a certain impact frequency. To achieve rotary motion with a certain impact frequency, the impact tool needs to include an assembly for outputting a rotary force and an impact mechanism for periodically impacting an output assembly. The impact mechanism can output rotary motion with a certain impact frequency to make the output torque large.

(4) This part provides background information related to the present application, which is not necessarily the existing art.

#### SUMMARY

(5) An impact tool includes a main shaft provided with a main shaft ball groove, where the main shaft ball groove includes a first ball groove that extends spirally about a main shaft axis and is concave on an outer surface and a second ball groove that extends spirally about the main shaft axis and is concave on the outer surface; and an impact block provided with an impact ball groove that mates with the main shaft ball groove to accommodate a rolling ball, where the impact ball groove includes a third ball groove that mates with the first ball groove to accommodate the rolling ball and a fourth ball groove that mates with the second ball groove to accommodate the rolling ball. The included angle  $\alpha$  between the first ball groove and the second ball groove is less than the included angle  $\beta$  between the third ball groove and the fourth ball groove.

(6) In some examples, the included angle  $\alpha$  between the first ball groove and the second ball groove is less than  $117^\circ$ .

(7) In some examples, an impact tool includes a housing; a motor including a drive shaft that rotates about a first axis, where the drive shaft optionally rotates in a first direction or a second direction; an output shaft for outputting torque, where the tightening torque of the output shaft to a workpiece is greater than or equal to 700 foot-pounds; and an impact assembly that provides an impact force to the output shaft.

(8) In some examples, the impact assembly includes a main shaft driven by the motor and rotating about a main shaft axis; an impact block supported on the main shaft and rotating integrally with the main shaft; a hammer anvil mating with the impact block and struck by the impact block; and a rolling ball connecting the main shaft to the impact block.

(9) In some examples, the included angle  $\alpha$  between the first ball groove and the second ball groove is greater than  $100^\circ$  and less than  $117^\circ$ .

(10) In some examples, the included angle  $\beta$  between the third ball groove and the fourth ball groove is greater than  $118^\circ$  and less than  $130^\circ$ .

(11) In some examples, when the impact block rotates in the first direction, the rolling ball moves in the first ball groove and the third ball groove; and when the impact block rotates in the second direction, the rolling ball moves in the second ball groove and the fourth ball groove.

(12) In some examples, the diameter of a portion of the main shaft with the main shaft ball groove is greater than and equal to 22 mm.

(13) In some examples, the impact assembly further includes an elastic element that provides a force for the impact block to approach the hammer anvil, and two ends of the elastic element are separately connected to an abutting surface of the main shaft and the impact block.

(14) In some examples, the coefficient of elasticity  $K$  of the elastic element is greater than or equal to 81 N/mm.

(15) In some examples, the impact block reciprocates forward and backward along the main shaft axis relative to the main shaft while rotating on the main shaft, the impact block includes a first position at the farthest end of forward movement of the impact block and a second position at the farthest end of backward movement of the impact block, and the impact block located at the first position is engaged with the hammer anvil.

(16) In some examples, when the impact block is located at the second position, the distance  $L2$  between an end of the impact block facing the abutting surface and the abutting surface is less than or equal to 4 mm.

(17) In some examples, the axial stroke H1 of the impact block on the main shaft is greater than or equal to 14 mm and less than or equal to 20 mm.

(18) In some examples, the motor includes a stator and a rotor, the drive shaft is formed on or connected to the rotor, the stator includes a stator core and coil windings disposed on the stator core, and the length of the stator core is less than 18 mm.

(19) In some examples, a displacement sensor that detects motion state information of a target part is further included, where the displacement sensor is disposed in an accommodation space; the displacement sensor is an eddy current sensor; and the target part is formed on or connected to the drive shaft, and the target part and the drive shaft move according to a preset rule.

(20) In some examples, a controller configured to control a working state of the motor according to the motion state information provided by the displacement sensor is further included, where the motion state information includes position information of the target part, and the controller is further configured to determine the position of the rotor according to the position information of the target part.

(21) In some examples, a battery pack is further included, where the battery pack supplies power to the motor.

(22) In some examples, a holding portion for holding a sleeve is formed on or connected to a front end of the output shaft, and the length L1 from a rear end of the housing to a front end of the holding portion is less than or equal to 210 mm.

(23) An impact tool includes a housing including a first housing and a second housing, where rear end surfaces of the first housing and the second housing define a rear end of the housing; and an output shaft for outputting power, where a holding portion for holding a sleeve is formed on or connected to a front end of the output shaft, and the tightening torque of the output shaft to a workpiece is greater than or equal to 700 foot-pounds. The length L1 from the rear end of the housing to a front end of the holding portion is less than or equal to 210 mm.

(24) In some examples, an impact tool includes a motor supported by at least a first housing and a second housing and including a drive shaft for outputting power; a direct current power supply that supplies power to the motor; and an impact assembly for providing an impact force to an output shaft. The impact assembly includes a main shaft driven by the motor and rotating about a main shaft axis; an impact block supported on the main shaft and rotating integrally with the main shaft; a hammer anvil mating with the impact block and struck by the impact block; and a rolling ball connecting the main shaft to the impact block, where the main shaft is provided with a main shaft ball groove, and the impact block is provided with an impact ball groove that mates with the main shaft ball groove to accommodate the rolling ball.

(25) In some examples, an impact tool includes a transmission assembly used for transmitting the power outputted by the drive shaft to the impact assembly and disposed between the motor and the impact assembly.

(26) An impact tool includes a housing; and an output shaft for outputting power, where the tightening torque of the output shaft to a workpiece is greater than or equal to 700 foot-pounds. The length L1 from a rear end of the housing to a front end of the output shaft is less than or equal to 210 mm.

(27) In some examples, an impact tool includes a motor accommodated in a housing and including a drive shaft for outputting power; and an impact assembly for providing an impact force to an output shaft. The impact assembly includes a main shaft driven by the motor and rotating about a main shaft axis; an impact block supported on the main shaft and rotating integrally with the main shaft; a hammer anvil mating with the impact block and struck by the impact block; and a rolling ball connecting the main shaft to the impact block, where the main shaft is provided with a main shaft ball groove, and the impact block is provided with an impact ball groove that mates with the main shaft ball groove to accommodate the rolling ball.

(28) In some examples, an impact tool includes a transmission assembly used for transmitting the

power outputted by the drive shaft to the impact assembly and disposed between the motor and the impact assembly.

(29) In some examples, the housing includes a barrel at least partially accommodating the motor; and a tail housing connected to a rear side of the barrel, where the tail housing holds a rear bearing for supporting a rear end of the drive shaft, and a rear side surface of the tail housing is defined as the rear end of the housing.

(30) In some examples, a battery pack is further included, where the battery pack supplies power to the motor.

(31) In some examples, the weight of the impact tool without the battery pack is defined as the bare weight of the impact tool, and the ratio of the tightening torque to the bare weight of the impact tool is greater than or equal to 152 foot-pounds per pound.

(32) A rotary power tool includes a power supply; a housing including a power supply coupling portion for connecting the power supply; a motor accommodated in the housing and including a drive shaft that rotates around a drive axis; an output shaft drivingly connected to the drive shaft and used for outputting power, where the output shaft rotates about an output axis; and a hanging assembly used for hanging the rotary power tool in a first state and including an opening for a suspended part to enter so that the hanging assembly is suspended on the suspended part. When the rotary power tool is in the first state, the included angle  $\gamma$  between an extension direction of the drive axis and a horizontal direction is less than or equal to  $45^\circ$ , and the output shaft is located on an upper side of the power supply coupling portion.

(33) In some examples, the hanging assembly includes a mounting seat and a hook, where the mounting seat is mounted on the housing, the hook is detachably connected to the mounting seat, and the hook is provided with an opening.

(34) In some examples, the housing includes a tail housing, a barrel, and a head housing connected in sequence, and the installation position of the mounting seat includes the barrel, the joint between the barrel and the head housing, the head housing, the joint between the barrel and the tail housing, or the tail housing.

(35) In some examples, two mounting seats are separately provided on two sides of the barrel, two sides of the joint between the barrel and the head housing, two sides of the head housing, two sides of the joint between the barrel and the tail housing, or two sides of the tail housing.

(36) In some examples, the mounting seat includes a first assembly portion, the hook includes a second assembly portion and a hook body, the second assembly portion is detachably connected to the first assembly portion, the hook body includes a storage state and a hanging state, and the hook body in the storage state is close to the housing.

(37) In some examples, a device accessory of the second assembly portion is selectively mounted on the mounting seat through the first assembly portion.

(38) In some examples, the device accessory includes a belt clip, a cord, or a bit clip.

(39) In some examples, the mounting seat includes connecting portions connected to the housing through fasteners.

(40) In some examples, the housing includes a barrel and a head housing, and the fastener simultaneously connects the barrel and the head housing to the mounting seat.

(41) In some examples, the housing includes a grip disposed on a lower side, and the hanging assembly is higher than the grip.

(42) A rotary power tool includes a power supply; a housing including a power supply coupling portion for connecting the power supply; a motor accommodated in the housing and including a drive shaft that rotates around a drive axis; an output shaft drivingly connected to the drive shaft and used for outputting power, where the output shaft rotates about an axis of the output shaft; and a hanging assembly used for hanging the rotary power tool in a first state and including a mounting seat and a body portion, where the mounting seat is mounted on a side of the housing. When the rotary power tool is in the first state, the output shaft is located on an upper side of the power

supply coupling portion.

(43) A rotary power tool includes a housing including a main housing and a grip for holding; a motor accommodated in the main housing and including a drive shaft that rotates about a drive axis; an output shaft drivingly connected to the drive shaft and used for outputting power, where the output shaft rotates about an output axis; a main switch for controlling the motor; and a first mounting portion for connecting a lanyard, where the first mounting portion is disposed at a position where the main housing and the grip are coupled, and the first mounting portion is close to a rear end of the main housing.

(44) In some examples, the first mounting portion is closer to the drive shaft relative to the main switch.

(45) In some examples, the main housing includes a barrel, a head housing, and a tail housing connected in sequence, and the grip is at least partially located below the barrel.

(46) In some examples, the housing includes a handle housing located on a lower side of the barrel, and the grip is located in the middle of the handle housing.

(47) In some examples, the rotary power tool includes a first mount, the first mount includes a first fixing portion and a first mounting portion, the first mounting portion includes a first lanyard hole, and the first mount is connected to the handle housing through the first fixing portion.

(48) In some examples, the handle housing includes a left handle housing and a right handle housing that are screwed together, and the first mount is sandwiched between the left handle housing and the right handle housing.

(49) In some examples, the tail housing is disposed at the rear end of the main housing, and along a direction of the drive axis, a rear end of the first mount does not extend beyond a rear end of the tail housing.

(50) In some examples, the first mount partially overlaps the barrel along a direction perpendicular to the drive axis.

(51) In some examples, the rotary power tool further includes a battery pack for supplying power to the motor, a power supply coupling portion is disposed at a lower part of the grip, and the power supply coupling portion is coupled with the battery pack.

(52) In some examples, the rotary power tool further includes a second mounting portion, a power supply coupling portion is disposed at a lower part of the grip, the second mounting portion is located on a rear side of the power supply coupling portion, and a lanyard is mounted on the first mounting portion or the second mounting portion.

(53) In some examples, the target part is a metal part.

(54) In some examples, the eddy current sensor includes a transmitting coil and a receiving coil, where the transmitting coil emits an alternating excitation signal to generate an alternating magnetic field during the operation of the motor, and the receiving coil receives an electrical signal generated by the movement of the target part in the alternating magnetic field and detects the position information of the target part according to the electrical signal.

(55) In some examples, the alternating excitation signal emitted by the transmitting coil is a sinusoidal signal, the electrical signal received by the receiving coil is a cosine signal, and the eddy current sensor determines the position information of the target part according to the sinusoidal signal and the cosine signal.

(56) In some examples, a first circuit board on which the transmitting coil and the receiving coil of the eddy current sensor are disposed and a second circuit board on which the controller is disposed are further included.

(57) In some examples, along the direction of the drive axis, the motor, the target part, and the first circuit board are arranged in sequence.

(58) In some examples, the first circuit board is fixed on an inner wall of the housing.

(59) In some examples, the eddy current sensor outputs a corresponding signal to the controller by demodulating and processing the received motion state information of the target part.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a structural view of an impact tool from a first angle according to an example of the present application;
- (2) FIG. 2 is a structural view of an impact tool from a second angle according to an example of the present application;
- (3) FIG. 3 is a sectional view of an impact tool according to an example of the present application, where an impact block is located at a first position;
- (4) FIG. 4 is a sectional view of an impact tool according to an example of the present application, where an impact block is located at a second position;
- (5) FIG. 5 is a structural view of an illumination element according to an example of the present application;
- (6) FIG. 6 is an exploded view of a housing according to an example of the present application;
- (7) FIG. 7 is an exploded view of an impact assembly according to an example of the present application;
- (8) FIG. 8 is a structural view of a main shaft according to an example of the present application;
- (9) FIG. 9 is an expansion view of a main shaft ball channel according to an example of the present application;
- (10) FIG. 10 is a structural view of an impact block according to an example of the present application;
- (11) FIG. 11 is an expansion view of an impact ball channel according to an example of the present application;
- (12) FIG. 12 is a schematic view illustrating the motion relationship between a main shaft ball channel, an impact ball channel, and a rolling ball according to an example of the present application and illustrates the total axial stroke of the impact block provided by a main shaft in FIG. 10;
- (13) FIG. 13 is an assembly schematic view of an output shaft and a main shaft according to an example of the present application;
- (14) FIG. 14 is a structural view of an impact tool and a suspended part according to an example of the present application;
- (15) FIG. 15 is an exploded view of a hanging assembly according to an example of the present application;
- (16) FIG. 16 is a structural view of a hook according to an example of the present application;
- (17) FIG. 17 is a structural view of a mounting seat according to an example of the present application;
- (18) FIG. 18 is a circuit diagram of an impact tool according to another example of the present application;
- (19) FIG. 19 is an exploded view illustrating part of the structure of an impact tool according to another example of the present application; and
- (20) FIG. 20 is a sectional view of an impact tool according to another example of the present application.

### DETAILED DESCRIPTION

- (21) Before any examples of this application are explained in detail, it is to be understood that this application is not limited to its application to the structural details and the arrangement of components set forth in the following description or illustrated in the above drawings.
- (22) In this application, the terms “comprising”, “including”, “having” or any other variation thereof are intended to cover an inclusive inclusion such that a process, method, article or device comprising a series of elements includes not only those series of elements, but also other elements

not expressly listed, or elements inherent in the process, method, article, or device. Without further limitations, an element defined by the phrase “comprising a . . .” does not preclude the presence of additional identical elements in the process, method, article, or device comprising that element.

(23) In this application, the term “and/or” is a kind of association relationship describing the relationship between associated objects, which means that there can be three kinds of relationships. For example, A and/or B can indicate that A exists alone, A and B exist simultaneously, and B exists alone. In addition, the character “/” in this application generally indicates that the contextual associated objects belong to an “and/or” relationship.

(24) In this application, the terms “connection”, “combination”, “coupling” and “installation” may be direct connection, combination, coupling or installation, and may also be indirect connection, combination, coupling or installation. Among them, for example, direct connection means that two members or assemblies are connected together without intermediaries, and indirect connection means that two members or assemblies are respectively connected with at least one intermediate members and the two members or assemblies are connected by the at least one intermediate members. In addition, “connection” and “coupling” are not limited to physical or mechanical connections or couplings, and may include electrical connections or couplings.

(25) In this application, it is to be understood by those skilled in the art that a relative term (such as “about”, “approximately”, and “substantially”) used in conjunction with quantity or condition includes a stated value and has a meaning dictated by the context. For example, the relative term includes at least a degree of error associated with the measurement of a particular value, a tolerance caused by manufacturing, assembly, and use associated with the particular value, and the like. Such relative term should also be considered as disclosing the range defined by the absolute values of the two endpoints. The relative term may refer to plus or minus of a certain percentage (such as 1%, 5%, 10%, or more) of an indicated value. A value that did not use the relative term should also be disclosed as a particular value with a tolerance. In addition, “substantially” when expressing a relative angular position relationship (for example, substantially parallel, substantially perpendicular), may refer to adding or subtracting a certain degree (such as 1 degree, 5 degrees, 10 degrees or more) to the indicated angle.

(26) In this application, those skilled in the art will understand that a function performed by an assembly may be performed by one assembly, multiple assemblies, one member, or multiple members. Likewise, a function performed by a member may be performed by one member, an assembly, or a combination of members.

(27) In this application, the terms “up”, “down”, “left”, “right”, “front”, and “rear” and other directional words are described based on the orientation or positional relationship shown in the drawings, and should not be understood as limitations to the examples of this application. In addition, in this context, it also needs to be understood that when it is mentioned that an element is connected “above” or “under” another element, it can not only be directly connected “above” or “under” the other element, but can also be indirectly connected “above” or “under” the other element through an intermediate element. It should also be understood that orientation words such as upper side, lower side, left side, right side, front side, and rear side do not only represent perfect orientations, but can also be understood as lateral orientations. For example, lower side may include directly below, bottom left, bottom right, front bottom, and rear bottom.

(28) In this application, the terms “controller”, “processor”, “central processor”, “CPU” and “MCU” are interchangeable. Where a unit “controller”, “processor”, “central processing”, “CPU”, or “MCU” is used to perform a specific function, the specific function may be implemented by a single aforementioned unit or a plurality of the aforementioned unit.

(29) In this application, the term “device”, “module” or “unit” may be implemented in the form of hardware or software to achieve specific functions.

(30) In this application, the terms “computing”, “judging”, “controlling”, “determining”, “recognizing” and the like refer to the operations and processes of a computer system or similar



electronic computing device (e.g., controller, processor, etc.).

(31) To clearly illustrate technical solutions of the present application, an upper side, a lower side, a left side, a right side, a front side, and a rear side are defined in the drawings of the specification.

(32) FIGS. **1** to **20** show a rotary power tool according to an example of the present application. As shown in FIGS. **1** and **2**, in this example, the rotary power tool is an impact tool. For example, the impact tool is an impact wrench **10**. It is to be understood that in other alternative examples, depending on the work accessory connected to the front end, the impact tool may be an impact drill, an impact screwdriver, or the like. In other alternative examples, the rotary power tool may be a drill, a screwdriver, or the like.

(33) In this example, the impact wrench **10** includes a power supply, a housing **100**, a motor **200**, an output shaft **400**, an impact assembly **300**, a transmission assembly **270**, and a hanging assembly **500**. The power supply is used for supplying power to the motor **200**. In this example, the power supply is a direct current power supply **800**. In this example, the direct current power supply **800** is a battery pack, and the battery pack mates with a corresponding power supply circuit to supply power to the electrical components in the impact wrench **10**. It is to be understood by those skilled in the art that the power supply is not limited to the scenario where the direct current power supply **800** is used, and the power may be supplied to the circuit elements through mains power, an alternating current power supply, or a combination of mains power and the battery pack in conjunction with the corresponding rectifier circuit, filter circuit, and voltage regulator circuit.

(34) In this example, the power supply is the battery pack **800**. In the following description, the power supply is replaced by the battery pack **800**, which is not intended to limit the present application. In this example, the battery pack **800** may be a lithium battery pack, a solid-state battery pack, or a pouch battery pack. In some examples, the nominal voltage of the battery pack **800** is greater than or equal to 18 V. In some examples, the nominal voltage of the battery pack **800** is 24 V or 36 V, where the weight of the impact wrench **10** without the battery pack **800** is defined as the bare weight of the impact wrench **10**.

(35) The motor **200** is at least partially accommodated in the housing **100**. The motor **200** includes a drive shaft **210** that rotates about a drive axis **101**. In this example, the motor **200** is specifically an inrunner. In the following description, the motor is replaced by the electric motor **200**, which is not intended to limit the present application.

(36) As shown in FIG. **3**, the electric motor **200** includes a stator **220** and a rotor **230**, and the drive shaft **210** is formed on or connected to the rotor **230**. The stator **220** includes a stator core and coil windings disposed on the stator core. Two ends of the drive shaft **210** extend from the rotor **230**. The front end of the drive shaft **210** is supported by a front bearing **240**, and the rear end of the drive shaft **210** is supported by a rear bearing **250**. In this example, the length of the stator core, that is, the stack length of the electric motor **200**, is less than 18 mm. The drive shaft **210** optionally rotates in a first direction or a second direction, where the first direction and the second direction are opposite. That is to say, the electric motor **200** optionally performs forward rotation and reverse rotation.

(37) As shown in FIGS. **1** to **6**, the housing **100** includes a body housing **110** and a handle housing **120**. The inner wall surface of the body housing **110** surrounds an accommodation space, and the electric motor **200**, the transmission assembly **270**, the output shaft **400**, and the impact assembly **300** are all at least partially disposed in the accommodation space. In this example, the body housing **110** includes a tail housing **113**, a barrel **111**, and a head housing **112** connected in sequence. That is to say, the body housing **110** has a cylindrical multi-section housing structure. In this example, the tail housing **113** holds the rear bearing **250** for supporting the rear end of the drive shaft **210**, and the rear side surface of the tail housing **113** is defined as the rear end of the housing **100**.

(38) The handle housing **120** further includes a grip **123** formed on or connected to the body housing **110**. A power supply coupling portion **124** for connecting the battery pack **800** is formed

on or connected to the grip **123**. The battery pack **800** is detachably connected to the power supply coupling portion **124**. In this example, the battery pack **800** is detachably connected to the power supply coupling portion **124**, that is, the battery pack **800** is detachably connected to the grip **123**. The handle housing **120** and the body housing **110** have a split structure and are connected through fasteners. In some examples, the power supply coupling portion **124** for connecting the battery pack **800** is formed on or connected to the handle housing **120**, and the battery pack **800** is detachably connected to the power supply coupling portion **124**.

(39) In some examples, the housing **100** includes a first housing and a second housing, and rear end surfaces of the first housing and the second housing define the rear end of the housing **100**. The electric motor **200** is supported by at least the first housing and the second housing. That is to say, the housing **100** includes a left half housing and a right half housing that can be spliced together, a front half housing and a rear half housing that can be spliced together, or an upper half housing and a lower half housing that can be spliced together.

(40) In some examples, the body housing **110** includes a first housing and a second housing, and rear end surfaces of the first housing and the second housing define the rear end of the housing **100**. The electric motor **200** is supported by at least the first housing and the second housing. That is to say, the body housing **110** includes a left half housing and a right half housing that can be spliced together, a front half housing and a rear half housing that can be spliced together, or an upper half housing and a lower half housing that can be spliced together. The handle housing **120** and the body housing **110** are connected to form the housing **100**.

(41) An output mechanism includes the output shaft **400** for connecting the work accessory and driving the work accessory to rotate. As shown in FIG. **13**, a holding portion **410** for holding a sleeve is formed on or connected to the front end of the output shaft **400**. In other alternative examples, a clamping portion is disposed at the front end of the output shaft **400** and can clamp corresponding work accessories, such as screwdrivers or drill bits, when implementing different functions.

(42) As shown in FIGS. **1** to **4**, the output shaft **400** is used for outputting power. The output shaft **400** rotates about an output axis **401**. In this example, the drive axis **101** coincides with the output axis **401**. In other alternative examples, the drive axis **101** and the output axis **401** are parallel to each other but do not coincide with each other. In other alternative examples, a certain angle exists between the drive axis **101** and the output axis **401**.

(43) As shown in FIG. **3**, the transmission assembly **270** is disposed between the electric motor **200** and the impact assembly **300** and used for transmitting power between the drive shaft **210** and the impact assembly **300**. In this example, the transmission assembly **270** is decelerated by a planet gear. The working principle according to which the planet gear performs the deceleration and the deceleration implemented by the transmission assembly **270** have been completely disclosed to those skilled in the art. Therefore, a detailed description is omitted herein for the brevity of the specification.

(44) As shown in FIGS. **3** and **4** and FIGS. **7** to **13**, the impact assembly **300** is used for providing an impact force to the output shaft **400**. The impact assembly **300** includes a main shaft **310**, an impact block **320** sleeved on the circumference of the main shaft **310**, a hammer anvil **330** disposed at the front end of the impact block **320**, and an elastic element **340**. The hammer anvil **330** is connected to the output shaft **400**. In this example, the hammer anvil **330** includes an anvil **331**, and the output shaft **400** is disposed at the front end of the anvil **331**. It is to be understood that the anvil **331** and the output shaft **400** may be integrally formed or separately formed as independent parts. The impact block **320** is driven by the main shaft **310**, and the hammer anvil **330** mates with the impact block **320** and is struck by the impact block **320**. A pair of radially symmetrical first end teeth **323** are convexly disposed on the front end surface of the impact block **320**. A pair of radially symmetrical second end teeth **332** are convexly disposed on the rear end surface of the anvil **331**.

(45) The elastic element **340** is disposed between the impact block **320** and an abutting surface **315**

of the main shaft **310** and used for providing a force for the impact block **320** to approach the hammer anvil **330**. In this example, the elastic element **340** is a coil spring.

(46) The impact assembly **300** further includes a rolling ball **350**. The rolling ball **350** connects the impact block **320** to the main shaft **310**. In this example, the rolling ball **350** is a steel ball. As shown in FIGS. 7 to 12, a main shaft ball groove **311** is formed on the outer surface of the main shaft **310**. An impact ball groove **321** that mates with the main shaft ball groove **311** to accommodate the rolling ball **350** is disposed on the impact block **320**. The main shaft ball groove **311** includes a first ball groove **3111** and a second ball groove **3112** that are spirally concave around a main shaft axis **301**. The impact ball groove **321** includes a third ball groove **3211** that mates with the first ball groove **3111** to accommodate the rolling ball **350** and a fourth ball groove **3212** that mates with the second ball groove **3112** to accommodate the rolling ball **350**. When the impact block **320** rotates in the first direction, that is, the forward rotation direction of the electric motor **200**, the rolling ball **350** moves in the first ball groove **3111** and the third ball groove **3211**. When the impact block **320** rotates in the second direction, that is, the reverse rotation direction of the electric motor **200**, the rolling ball **350** moves in the second ball groove **3112** and the fourth ball groove **3212**. The included angle  $\alpha$  between the first ball groove **3111** and the second ball groove **3112** is different from the included angle  $\beta$  between the third ball groove **3211** and the fourth ball groove **3212**. In this example, the included angle  $\alpha$  between the first ball groove **3111** and the second ball groove **3112** is less than the included angle  $\beta$  between the third ball groove **3211** and the fourth ball groove **3212**, and the included angle  $\alpha$  between the first ball groove **3111** and the second ball groove **3112** is less than  $117^\circ$ . The main shaft axis **301** coincides with the drive axis **101**. In other alternative examples, the drive axis **101** and the main shaft axis **301** are parallel to each other but do not coincide with each other.

(47) In some examples, the included angle  $\alpha$  between the first ball groove **3111** and the second ball groove **3112** is greater than  $100^\circ$  and less than  $117^\circ$ . In other examples,  $\alpha$  is greater than  $100^\circ$  and less than  $115^\circ$ . In some examples, the included angle  $\beta$  between the third ball groove **3211** and the fourth ball groove **3212** is greater than  $118^\circ$  and less than  $130^\circ$ . In other examples,  $\beta$  is greater than  $118^\circ$  and less than  $125^\circ$ .

(48) The main shaft ball groove **311** and the impact ball groove **321** both have semicircular groove bottoms. The rolling ball **350** straddles the impact ball groove **321** and the main shaft ball groove **311**. The impact ball groove **321** and the main shaft ball groove **311** jointly form a ball channel. The rolling ball **350** is disposed between the impact block **320** and the main shaft **310** and is embedded in the ball channel so that the main shaft **310** can drive the impact block **320** to rotate through the rolling ball **350**, and the impact block **320** mates with the hammer anvil **330** to drive the hammer anvil **330** to rotate, thereby further driving the output shaft **400** to rotate.

(49) In the related art, the impact block **320** is sleeved on the outer side of the main shaft **310**. Therefore, the diameter of a plane where the impact ball groove **321** is located is greater than the diameter of a plane where the main shaft ball groove **311** on the main shaft **310** is located. Moreover, when the main shaft axis **301** is used as a reference, the main shaft ball groove **311** disposed on the main shaft **310** is concave inward, which is equivalent to the following: the main shaft ball groove **311** is machined toward the main shaft axis **301**, and the impact ball groove **321** disposed on the impact block **320** is machined in a direction away from the main shaft axis **301**. The distance at which the rolling ball **350** moves in the main shaft ball groove **311** and the impact ball groove **321** is a function of the diameter of the part where the ball groove is located, the radius of the rolling ball **350**, and the included angle of the ball groove. In the related art, when the included angle  $\alpha$  between the first ball groove **3111** and the second ball groove **3112** is the same as the included angle  $\beta$  between the third ball groove **3211** and the fourth ball groove **3212**, the available length of the main shaft ball groove **311** is greater than the available length of the impact ball groove **321**, that is to say, the main shaft ball groove **311** is not fully utilized. In the present application, the included angle  $\alpha$  between the first ball groove **3111** and the second ball groove

**3112** is different from the included angle  $\beta$  between the third ball groove **3211** and the fourth ball groove **3212**. The included angle  $\alpha$  between the first ball groove **3111** and the second ball groove **3112** is less than the included angle  $\beta$  between the third ball groove **3211** and the fourth ball groove **3212**, which is equivalent to increasing the distance, that is, the length, of the impact ball groove **321**. The axial distance of the main shaft ball groove **311** along the main shaft axis **301** has a relatively large influence on the impact movement stroke of the impact block **320**. Therefore, the utilization rate of the main shaft ball groove **311** can be improved, which is more conducive to improving the output tightening torque of the impact wrench **10**. In this manner, the output stability and reliability of the product can be improved. In the present application, while the ball channel is fully utilized, the included angle  $\alpha$  between the first ball groove **3111** and the second ball groove **3112** is reasonably limited such that the relatively small included angle  $\alpha$  further increases the axial distance of the main shaft ball groove **311** along the main shaft axis **301** and increases the impact movement stroke of the impact block **320**.

(50) As shown in FIG. 12, **H1** denotes the total axial stroke of the impact block. The total axial stroke **H1** of the impact block is the sum of the axial distance at which the rolling ball **350** moves in the main shaft ball groove **311** and the axial distance at which the rolling ball **350** moves in the impact ball groove **321**. In this example, the total axial stroke **H1** of the impact block is less than or equal to 20 mm and greater than 14 mm. In some examples, the total axial stroke **H1** of the impact block is less than or equal to 19 mm and greater than 14 mm. In some examples, the total axial stroke **H1** of the impact block is less than or equal to 18 mm and greater than 14 mm. In some examples, the total axial stroke **H1** of the impact block is less than or equal to 17 mm and greater than 14 mm.

(51) In this example, the diameter of a portion of the main shaft **310** with the main shaft ball groove **311** is greater than and equal to 22 mm. In some examples, the diameter of a portion of the main shaft **310** with the main shaft ball groove **311** is greater than and equal to 23 mm.

(52) The diameter of the main shaft ball groove **311** of the main shaft **310** is increased so that the axial distance of the main shaft ball groove **311** along the main shaft axis **301** is further increased and the impact movement stroke of the impact block **320** is increased.

(53) In the preceding technical solutions, the utilization rate of the main shaft ball groove **311** of the main shaft **310** is improved without increasing the length of the main shaft **310**, and the included angle of the main shaft ball groove **311** and the included angle of the impact ball groove **321** are optimized so that the impact movement stroke of the impact block **320** is increased and the output tightening torque of the impact wrench **10** is increased.

(54) In this example, the helix angle of the first ball groove **3111** is equal to the helix angle of the second ball groove **3112**. In this manner, the main shaft **310** rotating at the same rotational speed in the first direction or the second direction has the same impact frequency.

(55) In the working process of the impact wrench **10**, the impact block **320** reciprocates forward and backward along the main shaft axis **301** of the main shaft **310** relative to the main shaft **310** while rotating on the main shaft **310**. The impact block **320** includes a first position at the farthest end of forward movement of the impact block **320** shown in FIG. 3 and a second position at the farthest end of backward movement of the impact block **320** shown in FIG. 4. The impact block **320** moving to the first position is engaged with the hammer anvil **330**. The distance **L2** between an end of the impact block **320** facing the abutting surface **315** and the abutting surface **315** is limited. As shown in FIG. 4, when the impact block **320** moves to the second position, the distance **L2** between the end of the impact block **320** facing the abutting surface **315** and the abutting surface **315** is less than or equal to 4 mm. In some examples, when the impact block **320** moves to the second position, the distance **L2** between the end of the impact block **320** facing the abutting surface **315** and the abutting surface **315** is less than or equal to 3 mm. In some examples, when the impact block **320** moves to the second position, the distance **L2** between the end of the impact block **320** facing the abutting surface **315** and the abutting surface **315** is less than or equal to 2

mm.

(56) In the related art, when the impact block **320** moves to the second position, the distance margin between the end of the impact block **320** facing the abutting surface **315** and the abutting surface **315** is relatively large. In this manner, the insufficient coefficient of elasticity  $K$  of the spring or the spring structural problem is avoided, and the spring is prevented from being compressed to too short and damaged. In the present application, the coefficient of elasticity  $K$  of the spring is greater than or equal to 81 N/mm so that the spring has sufficient capacity to resist the compression of the impact block **320**. Therefore, in the present application, when the impact block **320** moves to the second position, the distance  $L2$  between the end of the impact block **320** facing the abutting surface **315** and the abutting surface **315** is less than or equal to 4 mm. According to the relevant product dimensions, the axial length can be shortened by 6 mm to 7 mm. In some examples, the coefficient of elasticity  $K$  of the spring is greater than or equal to 85 N/mm. In some examples, the coefficient of elasticity  $K$  of the spring is greater than or equal to 90 N/mm.

(57) As shown in FIGS. **3**, **8**, and **13**, a connecting shaft **312** is disposed at an end of the main shaft **310** facing the output shaft **400**, the outer circumference of the connecting shaft **312** is provided with a first annular groove **3121**, the output shaft **400** is provided with a connecting groove **440**, and the connecting shaft **312** is rotatably disposed in the connecting groove **440**. In this manner, it is ensured that the output shaft **400** is radially limited. The diameter of the connecting shaft **312** is less than the diameter of the main shaft **310**, and the diameter of the main shaft **310** is greater than the diameter of the connecting groove **440**, thereby preventing the output shaft **400** from moving backward. The main shaft **310** is provided with a second abutting surface **3122** that abuts against the hammer anvil **330**. The connecting shaft **312** and the second abutting surface **3122** form a shaft shoulder structure, and the diameter of the second abutting surface **3122** is greater than the diameter of the connecting shaft **312**. The second abutting surface **3122** is basically perpendicular to the main shaft axis **301**. In this example, as shown in FIG. **8**, the distance  $L3$  between the second abutting surface **3122** and the abutting surface **315** of the main shaft **310** is less than or equal to 60 mm. In some examples, the distance  $L3$  between the second abutting surface **3122** and the abutting surface **315** of the main shaft **310** is less than or equal to 59 mm, 58 mm, 57 mm, 56 mm, or 55 mm. Since when the impact block **320** moves to the second position, the distance  $L2$  between the end of the impact block **320** facing the abutting surface **315** and the abutting surface **315** is reduced, the axis length of the part of the main shaft **310** where the impact block **320** moves is reduced. In this manner, the overall axial length is reduced.

(58) In this example, as shown in FIGS. **3** to **10**, the impact block **320** is provided with a mounting groove **324** along the direction of the main shaft axis **301**, the elastic element **340** is partially located in the mounting groove **324**, an end of the elastic element **340** abuts against the groove bottom of the mounting groove **324**, and the other end of the elastic element **340** abuts against the abutting surface **315** of the main shaft **310**. In this manner, the length of the impact block **320** and the length of the elastic element **340** partially overlap, thereby reducing the overall axial length of the impact wrench **10**. When the impact block **320** is at the second position, the compression amount of the spring is the maximum. The mounting groove **324** is annular and surrounds the outer circumference of the impact ball groove **321**. The spring is partially located in the mounting groove **324** and abuts against the groove bottom of the mounting groove **324**. In this manner, a certain gap exists between the spring and the main shaft **310** to avoid interference. In other examples, multiple mounting grooves **324** may be provided. The multiple mounting grooves **324** are evenly distributed around the main shaft axis **301**, and one spring is disposed in each mounting groove **324**.

(59) In this example, at least two main shaft ball grooves **311** are provided, and the at least two main shaft ball grooves **311** are evenly distributed around the main shaft axis **301** on the outer circumference of the main shaft **310**. In this manner, power is transmitted between the main shaft **310** and the impact block **320** through two rolling balls **350**, thereby improving the stability of power transmission, which is conducive to increasing the impact strength of the impact block **320**.

and the torque of the output shaft **400**.

(60) As shown in FIGS. **3** to **10**, the impact block **320** is provided with a mounting channel **322**, and the mounting channel **322** is sleeved on the main shaft **310**. The opening of the mounting groove **324** faces the abutting surface **315** on the rear side. The impact block **320** is provided with two annular structures at the mounting groove **324**. The two annular structures may be defined as a socket ring **325** and a protective ring **326**. The inner sidewall of the socket ring **325** is used for being in contact with the main shaft **310**, the outer sidewall of the socket ring **325** is the inner sidewall of the mounting groove **324**, and the inner sidewall of the protective ring **326** is the outer sidewall of the mounting groove **324**. From the groove bottom of the mounting groove **324**, the length of the socket ring **325** is greater than the length of the protective ring **326**. In this manner, on the premise that the contact area between the impact block **320** and the main shaft **310** is ensured, the outer diameter of the end of the impact block **320** facing the abutting surface **315** can be reduced, thereby providing more possibilities for the layout of the impact wrench **10**.

(61) The output shaft **400** is provided with a connection groove **420** that connects with the connecting groove **440**. The outer circumference of the output shaft **400** is provided with a second annular groove **430**. A rotary protective sleeve **450** is sleeved on the output shaft **400**. A third annular groove **451** is disposed on the inner sidewall of the rotary protective sleeve **450**. The third annular groove **451** and the second annular groove **430** are opposite and form an oil-containing annular cavity.

(62) In the related art, for the impact wrench **10**, the output shaft **400** is conventionally designed with a vent, an end of the vent connects with the connection groove **420**, and the other end of the vent connects with the second annular groove **430**. However, for a wrench with large torque, the vent is the weak point, possibly causing the output shaft **400** to break. If the vent is removed, after lubricating oil is added to the output shaft **400**, a certain vapor lock may be generated during the assembly of the main shaft **310** and the output shaft **400**, resulting in difficult assembly. To solve this problem, in the present application, an axially penetrating first accommodation cavity **313** is disposed on the main shaft **310**, and a rubber column **3131** is mounted in the first accommodation cavity **313**. During the assembly of the main shaft **310** and the output shaft **400**, the rubber column **3131** can provide a certain axial movement stroke for air flow and avoid or reduce the vapor lock during assembly. After assembly, the rubber column **3131** prevents the grease at the rear end of the output shaft **400** and in the gearbox from flowing through the first accommodation cavity **313** and the vent passage. In this manner, the vent is removed, thereby avoiding the problem of weakness of the output shaft **400** caused by the vent.

(63) As shown in FIG. **3**, a second accommodation cavity **314** is disposed on the rear part of the main shaft **310** and used for avoiding a drive gear **211** disposed at the front end of the drive shaft **210**, so as to reduce the overall length.

(64) In this example, the tightening torque of the output shaft to a workpiece is greater than or equal to 700 foot-pounds. It is to be explained that the “tightening torque” is the torque applied to a fastener in the direction of tightening the workpiece. That is, the impact block **320** can output the continuous rotational impact to the workpiece through the output shaft **400** with the torque **T** greater than or equal to 700 foot-pounds. The length **L1** from the rear end of the housing **100**, that is, the rear end of the tail housing **113** to the front end of the holding portion **410** is less than or equal to 210 mm. In some examples, the length **L1** from the rear end of the housing **100** to the front end of the holding portion **410** is less than or equal to 205 mm. In some examples, the length **L1** from the rear end of the housing **100** to the front end of the holding portion **410** is less than or equal to 200 mm.

(65) In some examples, the tightening torque of the output shaft to the workpiece is greater than or equal to 750 foot-pounds, and the length **L1** from the rear end of the housing **100**, that is, the rear end of the tail housing **113** to the front end of the holding portion **410** is less than or equal to 210 mm. In some examples, the length **L1** from the rear end of the housing **100** to the front end of the

[illegible]

**400** is less than or equal to 205 mm. In some examples, the length **L1** from the rear end of the housing **100** to the front end of the output shaft **400** is less than or equal to 200 mm.

(73) In some examples, when the body housing **110** or the housing **100** is a spliced structure of the first housing and the second housing and the rear end surfaces of the first housing and the second housing define the rear end of the housing **100**, the tightening torque of the output shaft to the workpiece is greater than or equal to 900 foot-pounds, and the length **L1** from the rear end of the housing **100** to the front end of the output shaft **400** is less than or equal to 210 mm. In some examples, the length **L1** from the rear end of the housing **100** to the front end of the output shaft **400** is less than or equal to 205 mm. In some examples, the length **L1** from the rear end of the housing **100** to the front end of the output shaft **400** is less than or equal to 200 mm.

(74) In this example, the ratio of the tightening torque to the bare weight of the impact tool is greater than or equal to 152 foot-pounds per pound. For example, when the bare weight of the impact tool is 10 pounds, the tightening torque is greater than or equal to 1520 foot-pounds. In some examples, the ratio of the tightening torque to the bare weight of the impact tool is greater than or equal to 155 foot-pounds per pound. In some examples, the ratio of the tightening torque to the bare weight of the impact tool is greater than or equal to 160 foot-pounds per pound. In some examples, the ratio of the tightening torque to the bare weight of the impact tool is greater than or equal to 170 foot-pounds per pound.

(75) As shown in FIGS. **1** and **2** and FIGS. **14** to **17**, in this example, the impact wrench **10** further includes the hanging assembly **500**. In some examples, the rotary power tool includes the hanging assembly **500**. In this example, the hanging assembly **500** is used for hanging the impact wrench **10** in a first state. The hanging assembly **500** includes an opening for a suspended part **900** to enter so that the hanging assembly **500** is suspended on the suspended part **900**. In this example, the opening faces the power supply coupling portion **124**. When the impact wrench **10** is in the first state, the included angle  $\gamma$  between the extension direction of the drive axis **101** and the horizontal direction is less than or equal to  $45^\circ$ . In this example, the extension direction of a suspension axis **901** is basically horizontal, and the included angle  $\gamma$  may be understood as the included angle between the extension direction of the drive axis **101** and the extension direction of the suspension axis **901**. In some examples, the extension direction of the drive axis **101** is basically parallel to the horizontal direction. In some examples, the extension direction of the drive axis **101** is basically parallel to the suspension axis **901**. The output shaft **400** is located on the upper side of the power supply coupling portion **124**. That is, when the impact wrench **10** is in the first state, the output shaft **400** is at a basically upper position. It is to be explained that in the product, due to tolerances, manufacturing errors, or measurement-related errors, the extension direction of the drive axis **101** is not completely parallel to the horizontal direction or the extension direction of the suspension axis **901**. Therefore, being parallel or basically parallel here should be considered as disclosing a range defined by absolute values of two endpoints. A parallel or basically parallel arrangement may mean that the included angle between the extension direction of the drive axis **101** and the horizontal direction or the extension direction of the suspension axis **901** is  $0^\circ$  plus or minus a certain percentage (for example, 1%, 5%, 10%, or more).

(76) In the related art, a U-shaped hook with an upward opening is used to hang the impact wrench **10**, causing the impact wrench **10** to hang upside down, that is, the output shaft **400** faces downward. When a user needs to use the impact wrench **10** again, the user needs to flip the impact wrench **10**, that is, the user needs to move the impact wrench **10** upward to remove the impact wrench **10** from a hanging rod **910** of the suspended part **900** and then rotate the tool by  $180^\circ$  to make the output shaft **400** face upward. Such operations affect the working efficiency. Moreover, a heavier impact tool places greater operational demands on the user, and there is a risk of the tool falling during the flipping process.

(77) In this example, in the case where the impact wrench **10** is hung in the first state, when the user hangs the impact wrench **10** or fetches and uses the impact wrench **10** again, the user does not



need to flip the rotary power tool, and the user only needs to move the rotary power tool downward to place the rotary power tool on the hanging rod **910** or move the rotary power tool upward to remove the rotary power tool from the hanging rod **910** so that the impact wrench **10** is always kept or basically kept at the working position, thereby improving the working efficiency. In this manner, the impact wrench **10** is prevented from being flipped when the impact wrench **10** is used again so that the output shaft **400** changes from downward to horizontal, thereby reducing wrist movement, which is conducive to wrist health. It is to be explained that the “hanging” in the first state of the impact wrench **10** is different from the safety rope hanging during working at a high altitude. The “hanging” state here means that when a set of operations is completed and the user needs to rest or use another tool, the user needs to stably hang the impact wrench **10** on the suspended part **900**. Generally, the suspended part **900** is not provided with an opening for the hanging assembly **500** to enter, and the hanging assembly **500** needs to be provided with an opening for the suspended part **900** to enter. After being hung, the hanging assembly **500** can bear the weight of the impact wrench **10** and stabilize the center of gravity, thereby ensuring that the impact wrench **10** does not shake or fall. The suspension axis **901** generally refers to the axial axis of the suspended part **900** at the position of the hanging assembly **500**, that is, the direction in which the suspended part **900** extends beyond two ends of the hanging assembly **500**. The axial axis of the suspended part **900** is basically perpendicular to the opening of the hanging assembly **500**. In this example, the suspended part **900** is a rod-like object. In this example, the suspended part **900** is a strip-like object.

(78) It is to be understood that when the rotary power tool is an electric screwdriver or an electric drill and does not need to provide an impact force, the impact assembly **300** does not need to be provided between the transmission assembly **270** and the output shaft **400**, which does not affect the relevant structural content of the hanging assembly **500**.

(79) In this example, the hanging assembly **500** includes a mounting seat **510** and a hook **520**, where the mounting seat **510** is mounted on the housing **100**, and the hook **520** is detachably connected to the mounting seat **510**. In this manner, the hook **520** is more convenient to maintain and replace. The hook **520** is provided with an opening facing downward.

(80) To ensure that the impact wrench **10** in the first state can maintain balance, in this example, the body housing **110** of the housing **100** includes the tail housing **113**, the barrel **111**, and the head housing **112** connected in sequence, and the installation position of the mounting seat **510** includes the barrel **111**, the joint between the barrel **111** and the head housing **112**, the head housing **112**, the joint between the barrel **111** and the tail housing **113**, or the tail housing **113**. The specific position may be set according to the internal structure and the position of the center of gravity of the whole impact wrench **10**, as long as the impact wrench **10** hung in the first state can maintain balance.

(81) In actual applications, the user may be accustomed to using the left hand to hold the impact wrench **10** or may be accustomed to using the right hand to hold the impact wrench **10**, or due to other factors such as injury, the user may only be able to use the impact wrench **10** with the left hand or the right hand. To adapt to the preceding situations, in this example, two mounting seats **510** are separately provided on two sides of the barrel **111**, two sides of the joint between the barrel **111** and the head housing **112**, two sides of the head housing **112**, two sides of the joint between the barrel **111** and the tail housing **113**, or two sides of the tail housing **113**. The two sides in this example refer to the left and right sides. That is, one mounting seat **510** is disposed on the left side of the barrel **111**, and one mounting seat **510** is disposed on the right side of the barrel **111**.

Alternatively, one mounting seat **510** is disposed on the left side of the joint between the barrel **111** and the head housing **112**, and one mounting seat **510** is disposed on the right side of the joint between the barrel **111** and the head housing **112**. Alternatively, one mounting seat **510** is disposed on the left side of the head housing **112**, and one mounting seat **510** is disposed on the right side of the head housing **112**. Alternatively, one mounting seat **510** is disposed on the left side of the tail housing **113**, and one mounting seat **510** is disposed on the right side of the tail housing **113**.

Alternatively, one mounting seat **510** is disposed on the left side of the joint between the barrel **111**

and the tail housing **113**, and one mounting seat **510** is disposed on the right side of the joint between the barrel **111** and the tail housing **113**.

(82) The hook **520** of the traditional impact wrench **10** is disposed at the power supply coupling portion **124**, is larger in dimension, and occupies a larger space. Therefore, a freely stowable hook **520** needs to be designed so that not only can the impact wrench **10** be hung on the hook **520**, but also the hook **520** is stowable and occupies a smaller space when the hook **520** is not in use.

(83) In this example, the mounting seat **510** is disposed on the body housing **110** of the housing **100** without affecting the use of the power supply coupling portion **124**. Therefore, in the bare state, the impact wrench **10** can stand.

(84) As shown in FIGS. **15** to **17**, the mounting seat **510** includes a first assembly portion **511**, the hook **520** includes a second assembly portion **521** and a hook body **522**, the second assembly portion **521** is detachably connected to the first assembly portion **511**, and the hook body **522** includes a storage state and a hanging state. The hook body **522** in the storage state may be close to the housing **100** or directly fit the housing **100**. The hook body **522** in the hanging state can be hooked on an object such as the hanging rod **910** of the suspended part **900**. That is to say, when the hook body **522** is in the storage state, a plane where the hook body **522** is located is parallel to the drive axis **101**. When the hook body **522** is in the hanging state, the plane where the hook body **522** is located is perpendicular to the drive axis **101**. That is, the hook body **522** needs to be rotated by approximately  $90^\circ$  between the storage state and the hanging state. In this example, the hook body **522** may have a flat plate structure, and the hook body **522** includes a flat plate part with an opening. In some examples, the hook body **522** is formed by bending a rod-like part into an opening in the same plane.

(85) Regarding the structure of the hook **520**, in some examples, the second assembly portion **521** is provided with a rotation hole, and the hook body **522** includes a first connecting rod **5221**, a second connecting rod **5222**, and a third connecting rod **5223**, where an end of the first connecting rod **5221** is connected to an end of the second connecting rod **5222**, the other end of the second connecting rod **5222** is connected to an end of the third connecting rod **5223**, the hook body **522** is in an inverted U shape as a whole, and the first connecting rod **5221** is rotatably connected to the rotation hole. When the hook body **522** is in the storage state, the second connecting rod **5222** is parallel to the drive axis **101**. When the hook body **522** is in the hanging state, the second connecting rod **5222** is perpendicular to the drive axis **101**. The first connecting rod **5221** is provided with a first stop portion **5225** and a second stop portion **5226**, where the first stop portion **5225** is located at an end of the rotation hole, and the second stop portion **5226** is located at the other end of the rotation hole. In this example, the first stop portion **5225** is located at the lower end of the rotation hole, and the second stop portion **5226** is located at the upper end of the rotation hole. In this manner, the first connecting rod **5221** is prevented from being pulled away from the rotation hole. A bending rod **5224** is disposed between the first connecting rod **5221** and the second connecting rod **5222**. When the hook body **522** is in the storage state, the bending rod **5224** is bent toward the housing so that the second connecting rod **5222** fits the housing **100**. The third connecting rod **5223** includes an arc-shaped rod and a vertical rod that are connected, and the other end of the second connecting rod **5222** is connected to the arc-shaped rod. When the hook body **522** is in the storage state, the arc-shaped rod fits the housing **100**, the curvature of the arc-shaped rod matches the diameter of the housing **100**, the vertical rod faces vertically downward, and the extension direction of the vertical rod is parallel to the extension direction of the first connecting rod **5221**. In this example, the first connecting rod **5221** extends in the up and down direction. The arrangement of the arc-shaped rod is conducive to saving the space occupied by the hook body **522** in the storage state, and the arrangement of the vertical rod is conducive to improving the hooking efficiency. When the hook body **522** is in the storage state, the arc-shaped rod is located at the rear side of the first connecting rod **5221**.

(86) To keep the hook body **522** in the storage state and the hanging state, in this example, a

position fixing structure is disposed between the second assembly portion **521** and the hook body **522**.

(87) Regarding the specific implementation of the position fixing structure, in an example, the first stop portion **5225** includes a stop pin, the diameter of the stop pin can be reduced, the axis of the rotation hole extends in the up and down direction, the second assembly portion **521** is provided with at least two intersecting limiting grooves **5211** at an end where the rotation hole is in contact with the stop pin, the first connecting rod **5221** is provided with a connecting rod hole, the axis extension direction of the connecting rod hole is perpendicular to the up and down direction, and the stop pin is disposed in the connecting rod hole; when the hook body **522** is in the storage state, the stop pin is located in one of the limiting grooves **5211**, and when the hook body **522** is in the hanging state, the stop pin is located in the other limiting groove **5211**. Regarding the diameter reducing structure of the stop pin, in this example, specifically, the stop pin is formed by rolling an elastic sheet and is in the shape of a column as a whole. The cross section of the stop pin is C-shaped. The stop pin with a C-shaped cross section facilitates diameter reduction. During operation, the first connecting rod **5221** rotates relative to the second assembly portion **521**. During rotation, the stop pin is pressed against the sidewall of the limiting groove **5211** so that the diameter of the stop pin becomes smaller, the distance between the stop pin and the second stop portion **5226** is increased, and the stop pin can cross the current limiting groove **5211** and enter the other limiting groove **5211**. In this example, the included angle between the extension directions of the two limiting grooves **5211** is  $90^\circ$ . In other examples, multiple limiting grooves **5211** may be provided so that the hook body **522** can be fixed at multiple angles when rotating relative to the second assembly portion **521**. In other examples, the stop pin may be formed by wrapping an elastic rubber cylinder around a cylindrical pin. The elastic rubber cylinder can deform when being pressed so that the diameter of the whole stop pin becomes smaller.

(88) Regarding the specific implementation of the position fixing structure, in another example, the first stop portion **5225** includes the stop pin, and the distance between the stop pin and the second stop portion **5226** can be changed. Specifically, the rotation hole extends in the up and down direction, the second assembly portion **521** is provided with at least two intersecting limiting grooves **5211** at an end where the rotation hole is in contact with the stop pin, the first connecting rod **5221** is provided with the connecting rod hole, the axis extension direction of the connecting rod hole is perpendicular to the up and down direction, the connecting rod hole is a long hole, the length direction of the connecting rod hole is the up and down direction, the stop pin is disposed in the connecting rod hole and is slidable along the up and down direction, and an elastic member is disposed between the second assembly portion **521** and the stop pin to make the stop pin approach the second stop portion **5226**; when the hook body **522** is in the storage state, the stop pin is located in one of the limiting grooves **5211**, and when the hook body **522** is in the hanging state, the stop pin is located in the other limiting groove **5211**. During operation, the first connecting rod **5221** rotates relative to the second assembly portion **521**. During rotation, the stop pin and the limiting groove **5211** are pressed against each other and relatively displaced so that the distance between the stop pin and the second stop portion **5226** is increased, and the stop pin can cross the current limiting groove **5211**. At this time, the elastic member deforms and stores energy. When the stop pin enters the other limiting groove **5211**, under the action of the elastic member, the stop pin abuts against the groove bottom of the other limiting groove **5211**.

(89) The opening of the limiting groove **5211** faces away from the second stop portion **5226**. The second stop portion **5226** is a limiting protrusion disposed on the first connecting rod **5221**. The limiting protrusion is formed by squeezing the outer circumferential wall of the first connecting rod **5221**.

(90) In this example, the grip **123** is located on the lower side of the body housing **110**, and the hook **520** is higher than the grip **123**. In other words, the hook **520** does not extend to the grip **123**, so as to avoid affecting the user's grip. The handle housing **120** is disposed on the lower side of the

housing **100**, and the grip **123** is located in the middle of the handle housing **120**. The extension direction of the grip **123** intersects with the extension direction of the drive axis **101**, and the coupling portion is disposed at an end of the grip **123**.

(91) To expand the adaptability of the first assembly portion **511**, in this example, a device accessory of the second assembly portion **521** is selectively mounted on the mounting seat **510** through the first assembly portion **511**. The device accessory includes a belt clip, a cord, or a bit clip. The belt clip may be worn on the belt of the user so that the impact wrench **10** can be hung on the belt of the user. The other end of the cord may be tied to the user or other devices, so as to prevent the impact wrench **10** from falling. The bit clip can accommodate multiple different types of bits, making it easy for the user to replace bits. Both the device accessory and the hook **520** may be referred to as the body portion.

(92) In this example, the mounting seat **510** includes connecting portions **512**, where the connecting portions **512** and the first assembly portion **511** are connected and integrally formed, and the connecting portions **512** are connected to the housing **100** through fasteners. Specifically, the connecting portion **512** is provided with a connecting hole **5121**, and the fastener is threadedly connected to the housing **100** after passing through connecting holes **5121**. In this example, the fastener includes a locking bolt **130**.

(93) In this example, as shown in FIG. **6** and FIGS. **15** to **17**, the fasteners simultaneously connect the barrel **111** and the head housing **112** to the mounting seat **510**. First connecting protrusions **1111** are convexly provided on the outer circumference of the barrel **111**, and the first connecting protrusion **1111** is provided with a first screw hole. Head connecting protrusions **1121** are convexly disposed on the outer circumference of the head housing **112**, and the head connecting protrusion **1121** is provided with a head mounting hole. In the example where the fasteners simultaneously connect the barrel **111** and the head housing **112** to the mounting seat **510**, multiple examples exist. In the first connection manner, the locking bolt **130** passes through the connecting hole **5121** and the head mounting hole and then is threaded into the first screw hole. In the second connection manner, the connecting portion **512** is located between the first connecting protrusion **1111** and the head connecting protrusion **1121**, and the locking bolt **130** passes through the head mounting hole and the connecting hole **5121** in sequence and then is threaded into the first screw hole. In the third connection method, two connecting portions **512** are provided, one of the connecting portions **512** is located between the first connecting protrusion **1111** and the head connecting protrusion **1121**, the other connecting portion **512** is located at an end of the head connecting protrusion **1121** facing away from the first connecting protrusion **1111**, and the locking bolt **130** passes through the connecting hole **5121** of one of the connecting portions **512**, the head mounting hole, and the connecting hole **5121** of the other connecting portion **512** in sequence and then is threaded into the first screw hole.

(94) In this example, the connecting portions **512** are disposed at two ends of the first assembly portion **511**, the first assembly portion **511** is used for connecting the second assembly portion **521**, and the connecting portions **512** are threadedly connected to the housing **100** through the fasteners. Specifically, two connecting portions **512** are disposed at an end of the first assembly portion **511**, two connecting portions **512** are disposed at the other end of the first assembly portion **511**, at least three first connecting protrusions **1111** are convexly provided on the outer circumference of the barrel **111**, the first connecting protrusion **1111** is provided with the first screw hole, at least three head connecting protrusions **1121** are convexly provided on the outer circumference of the head housing **112**, the head connecting protrusion **1121** is provided with the head mounting hole, the head connecting protrusions **1121** are disposed in one-to-one correspondence with the first connecting protrusions **1111**, the first assembly portion **511** is located between two adjacent first connecting protrusions **1111**, that is, between two adjacent head connecting protrusions **1121**, and for the connection relationship between the connecting portions **512** and the housing **100**, reference may be made to the content in the third connection manner.

(95) In this example, the first assembly portion **511** is hinged with a connecting seat **5212** of the second assembly portion **521**. Specifically, the connecting seat **5212** is provided with a first hole **5213**, the first assembly portion **511** is provided with a bypass groove, one of two sidewalls of the bypass groove is provided with a second hole **5111**, the other one of the two sidewalls of the bypass groove is provided with a third hole **5112**, the connecting seat **5212** is located in the bypass groove, and a connecting pin **530** passes through the second hole **5111**, the first hole **5213**, and the third hole **5112**. Further, the third hole **5112** is a threaded hole, an end of the connecting pin **530** is provided with a stud, and the stud is threaded into the threaded hole. Furthermore, the outer diameter of the stud is less than the outer diameter of the connecting pin **530**. The other end of the connecting pin **530** is provided with a hexagon socket for matching a socket head cap screw. A limiting column **5214** is provided at an end of the connecting seat **5212** and partially fits the first assembly portion **511**, so as to prevent the connecting seat **5212** from rotating about the axis of the connecting pin **530**. In this example, the limiting column **5214** is located on a side of the first hole **5213**, the groove bottom of the bypass groove is provided with a through hole **5113**, and the limiting column **5214** can pass through the through hole **5113** and be sandwiched in the gap between the first assembly portion **511** and the housing **100**. The limiting column **5214** is located on the upper side of the first hole **5213**.

(96) The mounting seat **510** is formed by cutting and bending a sheet steel plate. The sheet steel plate is cut into an H shape, the through hole **5113** is disposed at the middle joint, two long sides are bent by 90°, the second hole **5111** is located on a long side, and the third hole **5112** is located on the other long side. The two long sides are separately the two sidewalls of the bypass groove. The connecting portion **512** is located at the end of the long side.

(97) As shown in FIGS. **1** to **6**, the impact wrench **10** further includes a main switch **260**. The main switch **260** is used for controlling the operation of the electric motor **200**, including the starting, stopping, and rotational speed of the electric motor **200**.

(98) A first mounting portion **541** is used for connecting a lanyard. The first mounting portion **541** is disposed at a position where the body housing **110** and the grip **123** are coupled, and the first mounting portion **541** is close to the rear end of the body housing **110**. The lanyard is connected to the first mounting portion **541** at this position so that when the rotary power tool is used, the lanyard is connected upward to a safety lever, and the lanyard is located above the grip **123** and behind the main switch **260** and does not block the main switch **260**, thereby improving the user experience. In the example with the handle housing **120**, the first mounting portion **541** is disposed at a position where the body housing **110** and the handle housing **120** are coupled.

(99) To improve the convenience of holding the impact wrench **10** again when the impact wrench **10** is connected to the safety rope, in this example, the first mounting portion **541** is closer to the drive shaft **210** than the main switch **260**. That is to say, when the impact wrench **10** is standing, the first mounting portion **541** is located on the upper side of the main switch **260**. The specific position of the first mounting portion **541** may be reasonably adjusted according to the mass distribution of the whole impact wrench **10**. When an operator holds the impact wrench **10**, the first mounting portion **541** is close to the pulicue, and then when the thumb and other fingers hold the grip **123**, the forefinger can just touch the main switch **260**. The lanyard is located above the hand of the operator, thereby improving the efficiency of accurate holding and greatly improving the user experience.

(100) For example, the lanyard is connected to the safety lever of the user at the work site so that if the user drops the impact wrench **10**, the lanyard, a first mount **540**, and the housing **100** cooperate to prevent the impact wrench **10** from hitting the ground.

(101) In this example, the grip **123** is at least partially located below the barrel **111**. The head housing **112** is disposed at the front end of the whole body housing **110**, and the rear end portion of the head housing **112** extends into the front end portion of the barrel **111**. The preceding arrangement can effectively improve the connection strength between the head housing **112** and the

barrel **111** and is conducive to improving the sealing performance.

(102) In an example in which the hanging assembly **500** is not provided, the barrel **111** and the head housing **112** are connected through threaded fasteners. Through the threaded fasteners, the barrel **111** and the head housing **112** are firmly connected and are easy to install. In this example, the first connecting protrusions **1111** are provided along the circumferential direction of the barrel **111**, the first connecting protrusion **1111** is provided with the first screw hole, the head connecting protrusions **1121** are provided on the head housing **112**, the head connecting protrusion **1121** is provided with the head mounting hole, the threaded fastener includes the locking bolt **130**, and the locking bolt **130** passes through the head mounting hole and then threadedly mates with the first screw hole. Four first screw holes and four head mounting holes are provided separately. In other examples, three first screw holes and three head mounting holes are provided separately, or five first screw holes and five head mounting holes are provided separately. In this example, the first connecting protrusions and the head connecting protrusions **1121** can function as stiffeners to improve the strength of the body housing **110**.

(103) To improve the sealing performance of the body housing **110**, in this example, a sealing member is disposed between the barrel **111** and the head housing **112**. Specifically, the head housing **112** is provided with an annular groove, the sealing member includes a sealing ring, and the sealing ring is disposed in the annular groove. The sealing ring is sandwiched between the head housing **112** and the barrel **111**.

(104) In this example, further, anti-loosening glue is applied between the barrel **111** and the head housing **112**. The anti-loosening glue is provided so that the connection between the barrel **111** and the head housing **112** is stronger and the sealing performance is better.

(105) In this example, the tail housing **113** is located at the rear end of the body housing **110** and is connected to the barrel **111** through threaded fasteners. Specifically, second connecting protrusions **1112** are provided along the circumferential direction of the barrel **111**, the second connecting protrusion **1112** is provided with a second screw hole, tail connecting protrusions **1131** are provided on the tail housing **113**, the tail connecting protrusion **1131** is provided with a tail mounting hole, the threaded fastener includes the locking bolt **130**, and the locking bolt **130** passes through the tail mounting hole and then threadedly mates with the second screw hole. Four second screw holes and four tail mounting holes are provided separately. In other examples, three second screw holes and three tail mounting holes are provided separately, or five second screw holes and five tail mounting holes are provided separately.

(106) In the first example, the electric motor **200** is partially located in the barrel **111**. In the second example, the output shaft **400** is partially disposed in the head housing **112**. In the third example, the electric motor **200** is partially located in the barrel **111**, and the output shaft **400** is partially disposed in the head housing **112**. The transmission assembly **270** is located in the barrel **111**.

(107) In this example, the impact wrench **10** includes the first mount **540**, the first mount **540** includes a first fixing portion **542** and a first lanyard hole, the first mounting portion **541** includes the first lanyard hole, the lanyard passes through the first lanyard hole, is knotted, and then is fixedly connected to the first mount **540**, and the first mount **540** is connected to the handle housing **120** through the first fixing portion **542**. Specifically, the first fixing portion **542** includes a fixing hole, and a screw passes through the fixing hole and is connected to the housing **100** through a threaded fixing member. In this example, the threaded fixing member is a screw hole disposed in the housing **100** or a fixing member with a screw hole disposed in the housing **100**.

(108) The extension direction of the first lanyard hole is the left and right direction so that the direction in which the impact wrench **10** is placed is fixed.

(109) As shown in FIG. **6**, the handle housing **120** is located below the barrel **111**. In this example, the handle housing **120** includes a left handle housing **121** and a right handle housing **122** that are connected through the threaded fastener. In this example, the first mount **540** is sandwiched between the left handle housing **121** and the right handle housing **122**, and the screw passes

through the left housing and the first lanyard hole in sequence and then is connected to a threaded hole of the right housing through the threaded fastener.

(110) To prevent the first mount **540** from interfering with the operator during use, the tail housing **113** is disposed at the rear end of the body housing **110**, and along the direction of the drive axis **101**, the rear end of the first mount **540** does not extend beyond the rear end of the tail housing **113**. The first mount **540** partially overlaps the barrel **111** along a direction perpendicular to the drive axis **101**.

(111) The impact wrench **10** further includes a second mounting portion **551** located on the rear side of the power supply coupling portion **124**, and the user may selectively mount the lanyard on the first mounting portion **541** or the second mounting portion **551**.

(112) The impact wrench **10** includes a second mount **550**, and the second mount **550** includes the second mounting portion **551**. The second mounting portion **551** has a rod-like structure. A gap remains between the second mounting portion **551** and the housing **100**. The safety rope passes through this gap and is tied to the second mounting portion **551**.

(113) As shown in FIGS. **1** to **6**, the impact wrench **10** further includes an illumination element **600** and a protective cover **630**, the illumination element **600** is disposed on the head housing **112** and is used for generating light for illumination, the protective cover **630** covers at least the front of the illumination element **600**, a portion of the protective cover **630** located in front of the illumination element **600** is made of light-transmissive material, and the protective cover **630** is detachably connected to the head housing **112**. The illumination element **600** includes an annular light plate **610**, and the annular light board **610** includes a substrate and light beads. The substrate and the light beads are integrally formed. In some examples, the substrate and the light beads may be separate parts. The head housing **112** is provided with a light mounting groove **1122**, and the illumination element **600** is located in the light mounting groove **1122**. The protective cover **630** is detachably connected to the head housing **112**. The protective cover **630** includes a transparent lampshade **640** and a soft rubber lampshade **650**. The soft rubber lampshade **650** is provided with an avoidance hole. The transparent lampshade **640** is inserted into the avoidance hole. The soft rubber lampshade **650** may be made of opaque material. The transparent lampshade **640** and the soft rubber lampshade **650** provided separately can block the portion of the illumination element **600** that does not emit light to improve the appearance. A limiting step is disposed at the front end of the avoidance hole, the transparent lampshade **640** is provided with a stop portion, and the stop portion is located at the rear side of the limiting step and abuts against the limiting step. The limiting step can limit the forward displacement of the transparent lampshade **640**, thereby improving the stability of the transparent lampshade **640**.

(114) Regarding the fixation of the protective cover **630** and the head housing **112**, in this example, the stop part is disposed on the head housing **112**, and the front end of the soft rubber lampshade **650** abuts against the stop part. In this example, the rear end of the soft rubber lampshade **650** abuts against the opening end of the light mounting groove **1122**. The rear end of the transparent lampshade **640** also abuts against the opening end of the light mounting groove **1122**. The head housing **112** is provided with an annular snap groove, and an annular snap catch **810** is snap-fit with the annular snap groove, so as to prevent the soft rubber lampshade **650** from moving forward in the axial direction. The annular snap catch **810** is an elastic steel-wire circlip structure.

(115) A wiring duct **1123** is disposed on the lower side of the head housing **112**. An end of the wiring duct **1123** connects with the light mounting groove **1122**. A wire **620** of the illumination element **600** enters the handle housing **120** through the wiring duct **1123**. In this example, the lower side of the head housing **112** is connected to the handle housing **120** through a threaded boss. The threaded boss at least partially extends into the handle housing **120**. The wiring duct **1123** passes through the threaded boss. The impact wrench **10** further includes a decorative cover **820** that covers the opening of the wiring duct **1123**. The decorative cover **820** is inserted into the opening of the wiring duct **1123**. In other examples, the decorative cover **820** and the head housing **112** may

be connected through the threaded fastener. In this example, the threaded fastener may be a screw. In the direction of the drive axis **101**, the front end surface of the decorative cover **820** abuts against the rear end surface of the soft rubber lampshade **650**. The soft rubber lampshade **650** can limit the position of the decorative cover **820**, thereby improving the convenience of installation. (116) FIGS. **18** to **20** show another example of the present application. The impact wrench **10** is provided with a displacement sensor and a target part formed on or connected to the drive shaft. Components with the same functions as those in the first example have the same reference numerals.

(117) In this example, as shown in FIGS. **19** and **20**, a target part **212** is formed on or connected to the drive shaft **210**, and the target part **212** and the drive shaft **210** move according to a preset rule. As shown in FIG. **18**, a displacement sensor **720** detects the motion state information of the target part **212**, and the displacement sensor **720** is disposed in the accommodation space of the housing **100** and mounted outside the electric motor **200**. The displacement sensor **720** is a non-contact linear sensor. A controller **710** is configured to control the working state of the electric motor **200** according to the motion state information provided by the displacement sensor **720**.

(118) In some examples, the electric motor **200** is a three-phase brushless motor including a rotor with a permanent magnet and three-phase stator windings U, V, and W that are commutated electronically. In some examples, the three-phase stator windings U, V, and W adopt a star connection. In other examples, the three-phase stator windings U, V, and W adopt a delta connection. However, it is to be understood that other types of brushless motors are also within the scope of the present disclosure. The brushless motor may include less than or more than three phases of windings.

(119) The drive shaft **210** is formed on or connected to the rotor **230**. The drive shaft **210** is connected to the transmission assembly **270** and can transmit the torque outputted by the electric motor **200** to the transmission assembly **270**.

(120) The target part **212** is formed on or connected to the drive shaft **210** and may be formed by multiple pieces of fan-blade-shaped metal. The drive shaft **210** rotates about the drive axis **101**. During rotation, the target part **212** rotates together with the drive shaft **210** according to the preset rule. For example, the target part **212** may rotate synchronously with the drive shaft **210** or may rotate at N times the rotational speed of the drive shaft **210**, where  $N > 0$ . Since the rotational speed of the drive shaft **210** is related to the rotational speed of the rotor **230**, when the target part **212** and the drive shaft rotate according to the preset rule, the rotational speed of the target part **212** is related to the rotational speed of the rotor **230** so that the rotational speed of the rotor **230** can be detected by detecting the rotational speed of the target part **212**.

(121) The displacement sensor **720** may be located in the housing **100**, mounted outside the electric motor **200**, and not in contact with assemblies on the electric motor **200**, that is, not in contact with the stator **220**, the rotor **230**, the drive shaft **210**, and the target part **212** of the electric motor **200**. The displacement sensor **720** may be a non-contact linear sensor and may be disposed near the target part **212** so that the displacement sensor **720** can detect the motion state information of the target part **212** and send the detected motion state information to the controller **710**. In this manner, the controller **710** can control the working state of the electric motor **200** according to the motion state information provided by the displacement sensor **720**.

(122) In this example, the displacement sensor **720** may be an eddy current sensor. The eddy current sensor is used as the displacement sensor **720** for detecting the motion state of the rotor **230**. Compared to a contact sensor, the eddy current sensor does not increase the load on the electric motor **200** and can provide very accurate data even in harsh environments. When the electric motor **200** is started in an overloading working condition or at a low rotational speed with a load, the current in the electric motor **200** is increased. When a magnetic encoder (for example, a Hall sensor) in the related art is used, since the current interferes with the magnetic field of the magnet, the sensor such as the magnetic encoder detects the motion state of the electric motor



abnormally. The eddy current sensor is used so that the current interference with the magnetic field can be avoided, and the following problem can be solved: the electric motor **200** is susceptible to current interference in a large-current condition, causing false protection and starting with the load. (123) The motion state information of the target part **212** detected by the displacement sensor **720** may be the position information of the target part **212** so that the controller **710** can determine the position of the rotor **230** according to the position information of the target part **212** and output a driving control signal to the electric motor **200** according to the current position of the rotor **230**, thereby relatively accurately controlling the electric motor **200**. The displacement sensor **720** may collect the position information of one of the metal fan blades of the target part **212** so that the controller **710** can determine the position of the rotor **230** according to the position information of one metal fan blade. Alternatively, the displacement sensor **720** may collect the position information of multiple metal fan blades of the target part **212**; in this case, the controller **710** determines the position of the rotor **230** in conjunction with the position information of the multiple metal fan blades.

(124) The target part **212** is connected to or formed on the drive shaft **210** of the electric motor **200**, and the target part **212** and the drive shaft **210** move according to the preset rule so that the motion state information of the rotor **230** in the electric motor **200** can be determined by detecting the motion state information of the target part **212**. The eddy current sensor is used as the displacement sensor **720** for detecting the motion state information of the target part **212** so that the motion state information of the rotor **230** of the electric motor **200** can be detected, the load of the electric motor **200** can be reduced, very accurate data can be provided even in harsh environments, the current interference with the magnetic field can be avoided, and the following problem can be solved: the electric motor **200** is susceptible to current interference in a large-current condition, causing false protection and starting with the load. In this manner, the electric motor **200** can be controlled more accurately, which is conducive to improving the user experience.

(125) For example, the eddy current sensor includes a transmitting coil and a receiving coil, where the transmitting coil emits an alternating excitation signal to generate an alternating magnetic field during the operation of the electric motor **200**, and the receiving coil receives an electrical signal generated by the movement of the target part **212** in the alternating magnetic field and detects the position information of the target part **212** according to the electrical signal. The transmitting coil can transmit the alternating excitation signal, the alternating excitation signal generates an alternating electromagnetic field in space, and the receiving coil can receive the signal generated by the alternating electromagnetic field. The target part **212** induces the eddy current under the action of the alternating electromagnetic field, and the eddy current generates a secondary electromagnetic signal field. When the eddy current sensor and the target part **212** move relative to each other, the signal received by the receiving coil of the eddy current sensor changes. By demodulating and processing the received signal, the eddy current sensor can acquire the relative position between the eddy current sensor and the target part **212**, that is, acquire the position information of the target part **212**. In this case, the eddy current sensor outputs a corresponding signal, and the controller **710** controls the operation of the electric motor **200** based on the signal provided by the eddy current sensor.

(126) In some examples, the eddy current sensor outputs a corresponding signal to the controller **710** by demodulating and processing the received motion state information of the target part **212**.

(127) For example, the alternating excitation signal emitted by the transmitting coil is a sinusoidal signal, and the electrical signal received by the receiving coil is a cosine signal. The eddy current sensor determines the position information of the target part **212** according to the sinusoidal signal and the cosine signal. The eddy current sensor can calculate the ratio of the outputted sinusoidal signal to the received cosine signal to acquire the tangent value and then directly obtain the corresponding arctangent function value through the table lookup method according to the tangent value. The arctangent function value is the position information (that is, the angle) of the target part

**212.** In this manner, the angle of the rotor **230** in the electric motor **200** can be further determined according to the preset rule between the target part **212** and the drive shaft **210**.

(128) In some examples, in conjunction with FIGS. **19** and **20**, the impact wrench **10** further includes a first circuit board **730**. In this example, the transmitting coil and the receiving coil of the eddy current sensor are disposed on the first circuit board **730**. The controller **710** is disposed on a separate circuit board. In some examples, the transmitting coil and the receiving coil of the eddy current sensor and the controller **710** are all disposed on the first circuit board **730**. In this manner, through the circuit design of the circuit board, the eddy current sensor can transmit the acquired position information to the controller **710** in the form of an electrical signal, thereby improving the reliability of information transmission.

(129) In some examples, along the direction of the drive axis **101** of the drive shaft **210**, the electric motor **200**, the target part **212**, and the first circuit board **730** are arranged in sequence. In this example, since the target part **212** is formed on or connected to the drive shaft **210**, when the electric motor **200**, the target part **212**, and the first circuit board **730** are arranged in sequence along the direction of the drive axis **101**, the first circuit board **730** and the target part **212** are both located on the drive axis **101** of the drive shaft **210** so that it is convenient for the eddy current sensor to collect the motion state information of the target part **212**. For example, along the direction of the drive axis **101** of the drive shaft **210**, the target part **212** may be located on a side of the electric motor **200** facing away from the output shaft **400**, that is, on a side of the rear bearing **250** (or the fan) of the electric motor **200** facing away from the output shaft **400**, and the target part **212** may be disposed at an end of the drive shaft **210** facing away from the output shaft **400**. The output shaft **400** is usually used as the front end of the tool. In this case, the first circuit board **730** may be disposed at the position of the tail housing **113** of the tool so that the first circuit board **730** is opposite to the target part **212**, and thus it is convenient for the eddy current sensor disposed on the first circuit board **730** to detect the motion state information of the target part **212**.

(130) For example, the first circuit board **730** may be fixed on the inner wall of the housing **100**. In this manner, the position of the eddy current sensor can be fixed so that the positions of the eddy current sensor and the target part **212** are relatively fixed, thereby facilitating relatively accurate detection of the motion state information of the target part **212**.

(131) A control circuit is further included. FIG. **18** is a circuit diagram of a control circuit according to an example of the present invention. As shown in FIG. **18**, the control circuit includes a driver circuit **740** and the controller **710**. The driver circuit **740** is electrically connected to the stator windings U, V, and W of the electric motor **200** and is used for transmitting the current from the direct current power supply **800** to the stator windings U, V, and W so as to drive the electric motor **200** to rotate. In an example, the driver circuit **740** includes multiple switching elements Q1, Q2, Q3, Q4, Q5, and Q6. A gate terminal of each switching element is electrically connected to the controller **710** and is used for receiving a control signal from the controller **710**. The drain or source of each switching element is connected to the stator windings U, V, and W of the electric motor **200**. The switching elements Q1 to Q6 receive control signals from the controller **710** to change respective conduction states, thereby changing the current loaded to the stator windings U, V, and W of the electric motor **200** by the direct current power supply **800**. In an example, the driver circuit **740** may be a three-phase bridge driver circuit including six controllable semiconductor power devices (such as field-effect transistors (FETs), bipolar junction transistors (BJTs), or insulated-gate bipolar transistors (IGBTs)). It is to be understood that the preceding switching elements may be any other types of solid-state switches such as the IGBTs or the BJTs.

(132) In this example, the controller **710** is used for controlling the electric motor **200**. As shown in FIG. **3**, the impact wrench **10** is provided with a second circuit board **750**. In some examples, the second circuit board **750** is a main circuit board. The controller **710** may be disposed on the second circuit board **750** or the first circuit board **730**. In this example, the first circuit board **730** and the second circuit board **750** include a printed circuit board (PCB) and a flexible printed circuit (FPC)

board. The controller **710** adopts a dedicated control chip, for example, a single-chip microcomputer or a microcontroller unit (MCU). The controller **710** may control the direction of rotation of the electric motor **200** according to a steering switching signal provided by a switching portion **261** and may control the rotational speed of the electric motor **200** according to a signal provided by the main switch **260**. On this basis, the rotational speed of the electric motor **200** can be further adjusted according to the motion state information provided by the displacement sensor **720**. Specifically, the control chip controls the switching elements in the driver circuit **740** to be turned on or off. In some examples, the controller **710** controls the ratio of the on time of a drive switch to the off time of the drive switch based on a pulse-width modulation (PWM) signal. It is to be noted that the control chip may be integrated into the controller **710** or may be disposed independently of the controller **710**. The structural relationship between a driver chip (a chip used for the integrated driver circuit **740**) and the controller **710** is not limited in this example.

(133) Based on the same inventive concept, the examples of the present invention further provide a rotary power tool. The rotary power tool includes a housing formed with an accommodation space; an output portion driven to output torque, where the output torque of the output portion is greater than or equal to 500 foot-pounds; the electric motor **200** disposed in the accommodation space and including the stator and the rotor **230**; the drive shaft **210** formed on or connected to the rotor **230**, where the electric motor **200** drives the output portion through the drive shaft **210**; the target part **212** formed on or connected to the drive shaft **210**, where the target part **212** moves synchronously with the drive shaft **210**; an eddy current sensor used for detecting the motion state information of the target part **212** and disposed on the outer side of the electric motor **200**; and the controller **710** configured to control the working state of the electric motor **200** according to the motion state information provided by the displacement sensor **720**. The eddy current sensor is used to detect the motion state information of the rotor **230** of the electric motor **200** so that the load of the electric motor **200** can be reduced, very accurate data can be provided even in harsh environments, and the following problem can be solved: the electric motor **200** is susceptible to current interference in a large-current condition, causing false protection and starting with the load.

(134) The basic principles, main features, and advantages of this application are shown and described above. It is to be understood by those skilled in the art that the aforementioned examples do not limit the present application in any form, and all technical solutions obtained through equivalent substitutions or equivalent transformations fall within the scope of the present application.

## Claims

1. An impact tool, comprising: a housing; a motor comprising a drive shaft that rotates about a first axis, where the drive shaft optionally rotates in a first direction or a second direction; an output shaft for outputting a tightening torque greater than or equal to 700 foot-pounds; and an impact assembly that provides an impact force to the output shaft; wherein the impact assembly comprises a main shaft driven by the motor and rotating about a main shaft axis, an impact block supported on the main shaft and rotating integrally with the main shaft, a hammer anvil mating with the impact block and struck by the impact block, and a rolling ball connecting the main shaft to the impact block, a main shaft ball groove is provided on the main shaft and comprises a first ball groove that extends spirally about the main shaft axis and is concave on an outer surface and a second ball groove that extends spirally about the main shaft axis and is concave on the outer surface, the impact block is provided with an impact ball groove that mates with the main shaft ball groove to accommodate the rolling ball, the impact ball groove comprises a third ball groove that mates with the first ball groove to accommodate the rolling ball and a fourth ball groove that mates with the second ball groove to accommodate the rolling ball, an included angle  $\alpha$  between the first ball groove and the second ball groove is less than an included angle  $\beta$  between the third ball groove

and the fourth ball groove, and the included angle  $\alpha$  between the first ball groove and the second ball groove is less than  $117^\circ$ .

2. The impact tool of claim 1, wherein the included angle  $\alpha$  between the first ball groove and the second ball groove is greater than  $100^\circ$  and less than  $117^\circ$ .

3. The impact tool of claim 1, wherein the included angle  $\beta$  between the third ball groove and the fourth ball groove is greater than  $118^\circ$  and less than  $130^\circ$ .

4. The impact tool of claim 1, wherein, when the impact block rotates in the first direction, the rolling ball moves in the first ball groove and the third ball groove, and, when the impact block rotates in the second direction, the rolling ball moves in the second ball groove and the fourth ball groove.

5. The impact tool of claim 1, wherein a diameter of a portion of the main shaft with the main shaft ball groove is greater than and equal to 22 mm.

6. The impact tool of claim 1, wherein the impact assembly further comprises an elastic element that provides a force for the impact block to approach the hammer anvil, and two ends of the elastic element are separately connected to an abutting surface of the main shaft and the impact block.

7. The impact tool of claim 6, wherein a coefficient of elasticity  $K$  of the elastic element is greater than or equal to 81 N/mm.

8. The impact tool of claim 6, wherein the impact block reciprocates forward and backward along the main shaft axis relative to the main shaft while rotating on the main shaft, the impact block comprises a first position at a farthest end of forward movement of the impact block and a second position at a farthest end of backward movement of the impact block, and the impact block located at the first position is engaged with the hammer anvil.

9. The impact tool of claim 8, wherein, when the impact block is located at the second position, a distance  $L2$  between an end of the impact block facing the abutting surface and the abutting surface is less than or equal to 4 mm.

10. The impact tool of claim 8, wherein an axial stroke  $H1$  of the impact block on the main shaft is greater than or equal to 14 mm and less than or equal to 20 mm.

11. The impact tool of claim 1, wherein the motor comprises a stator and a rotor, the drive shaft is formed on or connected to the rotor, the stator comprises a stator core and coil windings disposed on the stator core, and a length of the stator core is less than 18 mm.

12. The impact tool of claim 11, further comprising a displacement sensor that detects motion state information of a target part, wherein the displacement sensor is disposed in an accommodation space, the displacement sensor is an eddy current sensor, the target part is formed on or connected to the drive shaft, and the target part and the drive shaft move according to a preset rule.

13. The impact tool of claim 12, further comprising a controller configured to control a working state of the motor according to the motion state information provided by the displacement sensor, wherein the motion state information comprises position information of the target part, and the controller is further configured to determine a position of the rotor according to the position information of the target part.

14. The impact tool of claim 1, further comprising a battery pack, wherein the battery pack supplies power to the motor.

15. The impact tool of claim 1, wherein a holding portion for holding a sleeve is formed on or connected to a front end of the output shaft, and a length  $L1$  from a rear end of the housing to a front end of the holding portion is less than or equal to 210 mm.

16. An impact tool, comprising: a housing comprising a first housing and a second housing, wherein rear end surfaces of the first housing and the second housing define a rear end of the housing; a motor supported by at least the first housing and the second housing and comprising a drive shaft for outputting power; a direct current power supply that supplies power to the motor; an output shaft having a holding portion for holding a sleeve formed on or connected to a front end of the output shaft and outputting a tightening torque greater than or equal to 700 foot-pounds; an

impact assembly used for providing an impact force to the output shaft; and a transmission assembly used for transmitting the power outputted by the drive shaft to the impact assembly and disposed between the motor and the impact assembly; wherein the impact assembly comprises a main shaft driven by the motor and rotating about a main shaft axis, an impact block supported on the main shaft and rotating integrally with the main shaft, a hammer anvil mating with the impact block and struck by the impact block, and a rolling ball connecting the main shaft to the impact block, the main shaft is provided with a main shaft ball groove, the impact block is provided with an impact ball groove that mates with the main shaft ball groove to accommodate the rolling ball, and a length L1 from the rear end of the housing to a front end of the holding portion is less than or equal to 210 mm.

17. An impact tool, comprising: a housing; a motor accommodated in the housing and comprising a drive shaft for outputting power; an output shaft for outputting a tightening torque greater than or equal to 700 foot-pounds; an impact assembly used for providing an impact force to the output shaft; and a transmission assembly used for transmitting the power outputted by the drive shaft to the impact assembly and disposed between the motor and the impact assembly; wherein the impact assembly comprises a main shaft driven by the motor and rotating about a main shaft axis, an impact block supported on the main shaft and rotating integrally with the main shaft, a hammer anvil mating with the impact block and struck by the impact block, and a rolling ball connecting the main shaft to the impact block, the main shaft is provided with a main shaft ball groove, the impact block is provided with an impact ball groove that mates with the main shaft ball groove to accommodate the rolling ball, and a length L1 from a rear end of the housing to a front end of the output shaft is less than or equal to 210 mm.

18. The impact tool of claim 17, wherein the housing comprises a barrel at least partially accommodating the motor; and a tail housing connected to a rear side of the barrel, wherein the tail housing holds a rear bearing for supporting a rear end of the drive shaft, and a rear side surface of the tail housing is defined as the rear end of the housing.

19. The impact tool of claim 17, further comprising a battery pack, wherein the battery pack supplies power to the motor.

20. The impact tool of claim 19, wherein a weight of the impact tool without the battery pack is defined as a bare weight of the impact tool, and a ratio of the tightening torque to the bare weight of the impact tool is greater than or equal to 152 foot-pounds per pound.

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